MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES 6SF7 AND 12SF7 1/
a b

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Diode, RF remote cutoff pentode

Outline a, b - - - 8-1 (EIA) Ⓐ

Base a, b - - - B8-21

Envelope a, b - - - MT8

Cathode a, b - - - Coated unipotential

Base connections:

Pin No.	a, b	1	2	3	4	5	6	7	8
Element	a, b	sh	g1	g3	g2	lp (diode)	p (pentode)	h	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	a,b	Ef	Eb	E1b	Ec1	Ec2	Pp	Pg2	I1b	Ehk Ⓐ	Alt
Unit:	a,b	V	Vdc	Vdc	Vdc	Vdc	W	W	mAdc	v	ft
Maximum:	a	6.9	330	---	---	110	3.5	0.5	1.0	100	10,000
Minimum:	a	5.7	---	---	---	---	---	---	---	---	---
Maximum:	b	13.9	330	---	---	110	3.5	0.5	1.0	100	10,000
Minimum:	b	11.3	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	a	6.3	250	0	-3	125	---	---	---	---	---
	b	12.6	250	0	-3	125	---	---	---	---	---

GENERAL:

Qualification - Required

1/ To identify those tests that are applicable to a given type or to several types; tube types are designated by letters.

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1266	Total grid current	a, b	See note 1	0.65	II	Ic1	0	-1.0	µAdc
1256	Electrode current (anode)	a, b		0.65	II	Ib	10.2	16.8	mAdc
1306	Transconductance (1)	a, b		0.65	II	Sm	1,650	2,450	µmhos
1231	Emission (diode)	a, b	Elb = 10 Vdc	0.65	II	Ilo	0.8	---	mAdc
1231	Emission (pentode)	a, b	Eb = Ec1 = Ec2 = 30 Vdc (see note 1)	0.65	II	Is	40	---	mAdc
Ⓐ 1201	Short and discontinuity detection	a, b		0.4	II	---	---	---	---
Ⓐ Appendix D, 30(a), 40	Visual and mechanical inspection criteria	a, b		---	---	---	---	---	---
Ⓐ Appendix D, 30(b)	Dimensions	a, b	Outline 8-1; dimensions A, B C, D, L, and M	1.0	I	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	a, b	Rp = 10,000 ohms	6.5	S3	Ep	---	200	mVac
1306	Transconductance (2)	a, b	Ec1 = -45 Vdc	6.5	S3	Sm	1	30	µmhos
1301	Heater current	a		6.5	S3	If	275	325	mA
		b		6.5	S3	If	138	162	mA
Ⓐ 1336	Heater-cathode leakage	a, b		6.5	S3	Ihk	---	20	µAdc
1331	Direct-interelectrode capacitance	a, b		6.5	S3	Cg1p Cg1k Cpk Cg11p C1pp	--- 4.4 4.2 --- 0.5	0.004 6.6 7.8 0.002 1.1	pF pF pF pF pF
1231	Electrode current (screen grid)	a, b		6.5	S3	Ic2	2.4	4.8	mAdc
Ⓐ 1101	Secureness of base, cap, or insert	a, b		6.5	S3	---	---	---	---
Ⓐ 1111	Base pin solder depth	a, b		6.5	S3	---	---	---	---
Ⓐ 1211	Insulation of electrodes	a, b		4.0	S3	R	10	---	Meg
Ⓐ 1105	Permanence of marking	a, b		---	---	---	---	---	---

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METHOD	REQUIREMENT OF TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3								
1501	Life-test provisions (intermittent) (500 hours)	a, b	Group A: I_{fb} = 1.0 mAdc E_{bk} = 100 V	---	---	---	---	---	---
---	Life-test end points (intermittent) (500 hours)	a, b	Transconductance (1)	---	---	g_m	1,525	---	µmhos
			Emission (diode)	---	---	I_{is}	0.3	---	mAdc

NOTE:

1. This test to be performed at the conclusion of the holding period.

The margins of this tube specification sheet are marked with a revision letter (A) to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Air Force - 11

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